

1 **Mechanical and electrical properties of concrete incorporating an iron-particle**
2 **contained nano-graphite by-product**

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Abstract

Electrically conductive cementitious composites (ECCC) have a wide range of potential applications in advanced structural technologies such as structural health monitoring and pavement deicing. General conductive fillers, such as steel fiber, carbon nanofiber and carbon nanotube, have been investigated to enhance the ECCC conductivity. However, cheaper and more efficient conductive fillers are required to promote the wider application of ECCC. The present study applies an iron-particle contained composite nano-graphite by-product (ING) to investigate a new type of conductive concrete. Both mechanical and electrical properties of this conductive concrete are examined and compared with corresponding measured properties of plain concrete and composites incorporating pure nano-graphite (NG). Results show a loss in compressive strength and increase in conductivity of concrete due to the addition of nano-graphite materials. A 1 wt% content of ING leads to almost 20% reduction in concrete's compressive strength compared with the plain concrete, which is larger than NG (only 6.3%). However, the mechanical behavior of concrete with ING is little better than that of concrete with NG at 3 wt% and 5 wt%. ING is more significant in enhancing concrete conductivity than NG, especially at 1 wt% by resulting in a 70.3% decrease in electrical resistivity while only 11.3% resistivity reduction is observed for NG at the same concentration. Microstructural images show that composites incorporating ING material reach the percolation threshold of resistivity sooner. Meanwhile, ING particles show agglomerated state within the concrete matrix while NG tends to spread among it in the form of platelets.

Keywords: conductive concrete; composite nano-graphite; compressive strength; electrical resistivity; microstructural imaging

1 Introduction

Cementitious composites (CC) play a significant role in infrastructure constructions such as buildings, bridges and road pavements due to their superior mechanical properties, high manufacturing and cost efficiency and easy accessibility [1]. However, CC-based structures may suffer safety issues such as cracking and degradation under extreme environments or after long-term services, which may lead to severe hazards. Appropriate structural health monitoring (SHM) strategies are thus vital to ensure that concrete structures maintain their intended safety under demanding working environments over the service lifecycle. Techniques that have been studied and proposed for sensing the structural health status include optical fiber sensors [2-4], acoustic emission [5-7] and camera image processing [8-10].

However, these methods either work in a passive way or generate varying levels of impact on the mechanical and physical properties of cementitious composites. Hence, the recent idea of smart structures using the concrete material itself as a sensor has drawn considerable attention [11]. Damage or structural changes to concrete structures can be recorded by detecting abnormal signals of the electrical resistivity during the measurement [12][13]. In order to improve the piezo-resistive effect of concrete, studies have focused on developing more efficient electrically conductive cementitious composites (ECCC) by understanding their mechanisms. New applications of ECCC

have also evolved during this process. For example, electrochemical extraction of chloride can be achieved by spraying the graphite-based cement paste upon the surface of reinforced concrete elements as a node [14]. ECCC is also central to the electrical heater system of pavements, which are used to remove accumulated snow; this innovation greatly reduces the degree of surface damage compared with traditional methods [15-18].

Undoubtedly, the incorporation of conductive fillers is the most efficient method of improving the electrical conductivity of cementitious composites, which are generally insulated with high electrical resistivities ranging from $10^6 \Omega \cdot \text{cm}$ to $10^9 \Omega \cdot \text{cm}$ [19]. A thorough review of all potential functional fillers and their performance in ECCCs has been conducted by Wang et al. [20]. Research on the use of conductive fillers in cementitious composites started with metallic materials, such as steel fiber, steel shaving and steel slag [21-22]. Steel fiber was proven to be more effective than the other two metallic materials in reducing concrete's electrical resistivity, while also enhancing the flexural and tensile strength due to the pull-out effect [23]. However, the corrosion behavior of metallic materials may affect the stability of conductive concrete's piezo-resistive effect. The incorporation of metallic particles in millimeter scale may also generate microcracks in the concrete matrix that will reduce the mechanical properties [24].

Therefore, carbon-based materials are now considered as the mainstream addition agents for creating conductive concrete due to the superior electrical conductivity and working stability; these include carbon fiber, carbon black, graphite powders as well as their formations at nanoscale. An inconsistent percolation threshold was noticed for the electrical resistivity of carbon black incorporated composites at different mixing proportions [25-26]. Sun et al. [27] found that the fractural change in electrical resistivity reaches 15.6% for cementitious composites filled with 2 vol% of multi-layer graphene (MLGs). Mo [28] proposed a percolation model that could predict the formation and development of conductive networks in graphite-modified concrete. Dong et al. [29] have worked on a novel cement-based sensor embedded with layer-distributed conductive rubber fibers collected from household rubbish like cables and achieved satisfactory results.

Compared with carbon-based materials at normal scale, nanomaterials become more popular because of their higher conductivity and higher packing density achievable through their use in cementitious composites [30]. In addition, carbon nanotubes and their associated high aspect ratios with strong van der Waals forces enhance the electrical and physical properties of cement composites [31]. The incorporation of both graphite nanoplatelets and carbon nanofibers can reduce the total porosity of ultra-high-performance concrete (UHPC) and increase the autogenous shrinkage at all ages [32][33]. The piezo-resistive effect of cementitious composites filled with nano-graphite platelets (NGPs) was investigated systematically under different loading conditions and experimental results reveal that 5 vol% of NGPs have the most sensitive piezo-resistivity among all composites [34-35]. Several causes for the fluctuation of electrical properties of cement composites incorporating carbon-based nanomaterials (CNMs) were identified and analyzed by Lee et al. [36] including

moisture contents and dispersion of CNMs.

The barriers to the wider use of carbon nanomaterials are their high cost and the difficulty in achieving adequate dispersion at large dosages. Cheaper substitutes with a similar or better conductive effect are thus required. Therefore, the present study examines a new type of conductive concrete (INGC) that employs an iron-particle contained nano-graphite material. This new impure nano-graphite material is a composite nano-graphite by-product generated during the hydrogen production process [37] that contains a high percentage of iron particles. An experimental study is described that investigates the mechanical and electrical properties of the proposed INGC compared with the concrete incorporating the normal nano-graphite (NGC). The contents of nano-graphite materials examined are 1%, 3% and 5% by the mass of cement. Compressive strength and electrical resistivity tests as well as the Scanning Electron Microscope (SEM) analysis are performed on concrete specimens of all mixtures. The study provides the basis for assessing the potential of using a cheaper composite nano-graphite by-product for producing conductive concrete.

2 Experimental procedure

2.1 Materials and mix design

Plain concrete (PC) with a design compressive strength of 42.5 MPa was used as the control material that comprised general Portland cement, fine aggregate and tap water. The general Portland cement employed was produced with the mixture of Portland clinker ($\approx 97\%$) and other additional minerals, and conforms to the Australia standard AS3972 [38]. Aggregates composed of general sand in AFS 40-50 and gravel with the maximum size of 4 mm were used as fillers. The concrete composites that contained nano-graphite materials were designed based on the composite proportion of the plain concrete by adding nano-graphite aqueous solution. In order to secure the uniform dispersion of nano-graphite particles in solution, a polycarboxylate-based superplasticizer was incorporated in all mixtures including the plain concrete. The density of this superplasticizer is approximately 1.08 kg/L with 4.3 ± 0.5 pH at 20 °C.

Two types of nano-graphite materials were investigated. One is the pure nano-graphite (NG) purchased from U.S. and the other is a composite nano-graphite containing iron particles (ING), which is a by-product collected from the hydrogen production process in Australia. Table 1 displays some basic information about these two nano-graphite materials. Obviously, the overall BET specific surface area of the pure nano-graphite is much larger than that of the composite nano-graphite. That is probably because the composite nano-graphite contains iron particles, which have greater mass and smaller surface area.

Table 1 Physical properties of nano-graphite materials

Type	BET (m ² /g)	Particle size distribution (μ m)	Cost (US\$/L)
Pure Nano-graphite (NG)	248.9	0.1-10	13.0
Composite Nano-graphite (ING)	13.0	0.1-18	-

2.2 Treatment of nanomaterials

For concrete composites mixed with nano-graphite materials, the nano-graphite liquid solution was prepared before the composite mixing. The preparation process

includes the following steps as depicted in Fig.1 [32-34]: (1) Superplasticizer and nano-graphite powder were added into a beaker that contained 150 g tap water successively; (2) The beaker was then placed inside the proof box of an ultrasonicator machine for dispersing nano-graphite particles evenly using the power of ultrasonic wave (the ultrasonicating parameters are listed in Table 2); A pause of 5 seconds was used to prevent overheating of the liquid; (3) The nano-graphite solution was finally placed aside until it cooled down to room temperature (21 ± 2 °C).

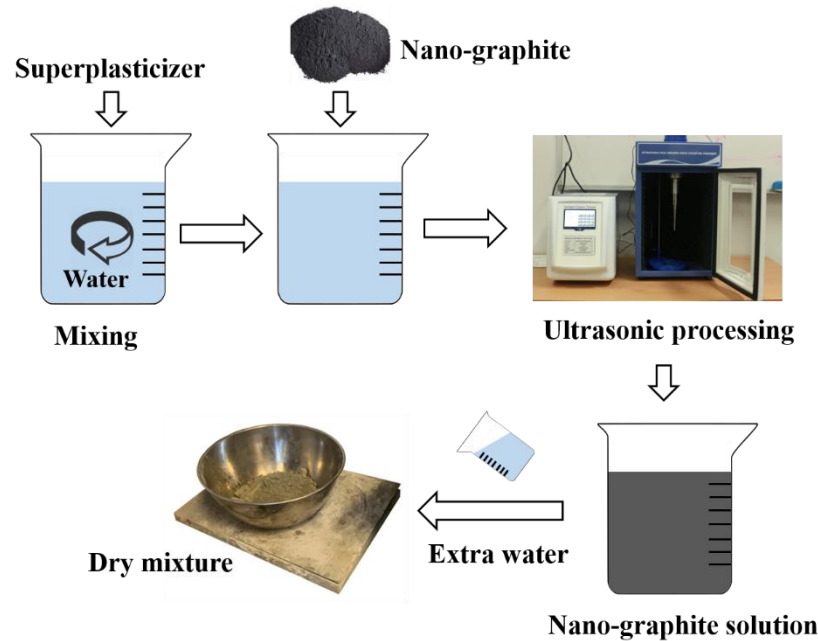


Fig.1 Schematic of nano-graphite involved concrete composite preparation

Table 2 Parameters of ultrasonic processing

Working Time	Energy	Pulse On	Pulse off	Horn Size
8 mins	40%	2 s	5 s	10 mm

2.3 Specimen preparation

The specimen preparation was performed in a laboratory environment (21 ± 2 °C). After weighing, all binders and aggregates were homogenized at 60 revs/min mixing speed for at least 5 mins. Tap water mixed with superplasticizer was added to the dry mixture to produce the plain concrete composite while the nano-graphite aqueous solution was used for concrete composites containing nano-graphite materials (as shown in Fig.1) and followed by additional 5 mins mixing at the same rate.

During this process, it was observed that, although the w/c ratio could be held constant for different contents of the composite nano-graphite (ING), the w/c ratio needed to be raised a little to maintain the workability of concrete paste when the content of pure nano-graphite (NG) powder increased. This phenomenon occurred probably because of the greater BET specific surface area of NG material and its significant absorption effect for water and superplasticizer that has a controlling impact on the flowability and workability of concrete paste [32]. As there is an inverse relationship between w/c ratio and concrete compressive strength [39], in order to investigate the influence of different types and contents of nano-graphite materials

alone, a certain w/c ratio (0.43) for concrete mixtures was determined by adjusting the dosage of superplasticizer in following compressive strength and electrical resistivity tests. The final proportions of all concrete mixtures are listed in Table 3.

Table 3 Proportions of concrete composites

	Cement (kg/m ³)	Water (kg/m ³)	w/c	Fine aggregate (kg/m ³)	Coarse aggregate (kg/m ³)	Nano- graphite (wt%)	Superplasticizer (wt%)
PC ^a	466	200	0.43	610	1109	0	0.7
NGC ^b 1	466	200	0.43	610	1109	1	0.7
NGC3	466	256	0.55	610	1109	3	0.7
NGC5	466	298	0.64	610	1109	5	0.7
NGCF ^c 1	466	200	0.43	610	1109	1	0.7
NGCF3	466	200	0.43	610	1109	3	1.0
NGCF5	466	200	0.43	610	1109	5	1.3
INGC ^d 1	466	200	0.43	610	1109	1	0.7
INGC3	466	200	0.43	610	1109	3	0.7
INGC5	466	200	0.43	932	1109	5	0.7

a: PC represents the plain concrete; b: NGC represents the pure nano-graphite filled concrete with different w/c ratio; c: NGCF represents the pure nano-graphite filled concrete with fixed w/c ratio; d: INGC represents the composite nano-graphite filled concrete.

For each mixture, three specimens were cast in cylindrical moulds (50 mm × 100 mm) after the mixing. The cast specimens were left in the laboratory (21±2 °C) covered by wet burlap and plastic sheet, and then demolded after 24 hours. All specimens were cured in the standard moisture room (25±2 °C and 95% RH) for another 28 days [40]. Figure 2 shows some of samples at the curing stage.



Fig.2 Some of samples at the curing stage

2.4 Experimental methods

2.4.1 Electrical resistivity

A digital multimeter was used to measure the electrical resistivity of concrete samples by the four-probe method as shown in Fig.3 [41-43]. In this method, four circular bands of copper tape that served as electrodes along the length of specimen with equal spacing (20 mm) were bonded around the side surface perimeter of specimens as displayed in Fig.4. While two interior electrodes measured the electric potential, direct current was applied to pass through the two outer probes. The copper

bands allowed the current to penetrate the whole cross section of the region between two voltage contacts. To ensure continuous contact between electrodes and concrete surface and minimize the contact resistance influence, surface of each specimen was cleaned with a wet towel before bonding and silver epoxy was used along the copper wires. By measuring the current and electrical potential, the multimeter can detect the resistance and then calculate the electrical resistivity using Equation 1.

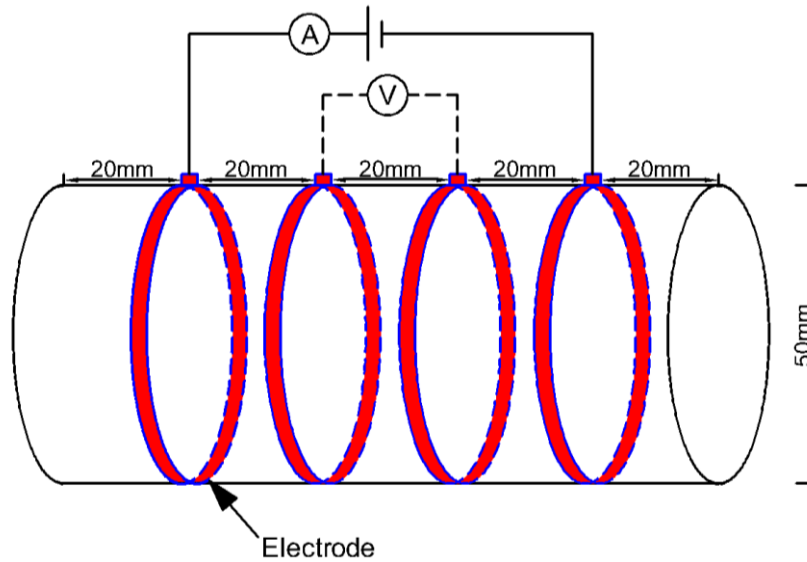


Fig.3 Schematic of four-probe electrical resistivity measurement

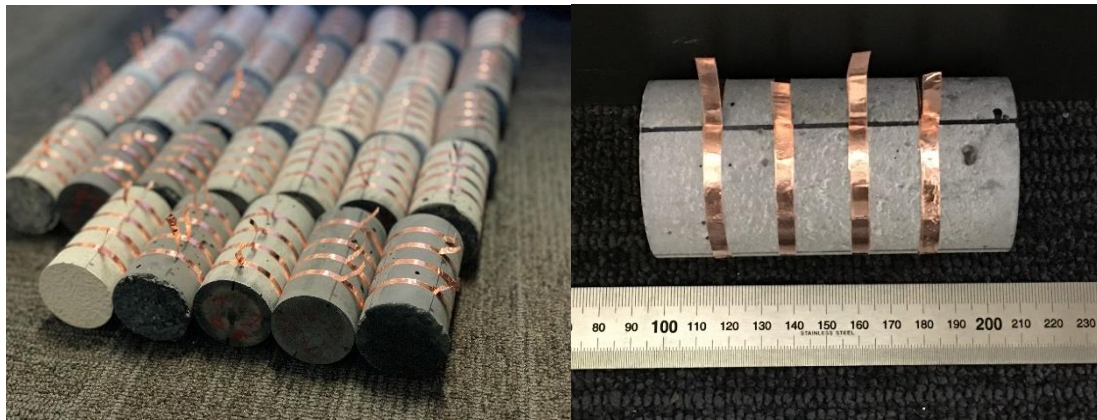


Fig.4 Images of copper tape bonded specimens

$$\rho = 2\pi a \frac{V}{I} \quad (1)$$

where ρ stands for the observed electrical resistivity; a denotes the distance between two close electrodes; V and I are the electric potential and current respectively.

2.4.2. Compressive strength

The compressive strength of concrete samples was tested using a uniaxial compressive machine with a loading capacity of 600 kN in accordance with the standard ASTM C 39 [44]. The loading rate was controlled by the air pressure system to ensure a constant loading procedure with 0.5 mm/min. This test was conducted at 28 days after the standard curing stage and directly followed the electrical resistivity test. Test data

including time, displacement and loading were saved by the computer software automatically. Pieces of debris for each mixture were collected after each sample's failure for later Scanning Electron Microscope (SEM) analysis.

2.4.3 Microstructure analysis

The Scanning Electron Microscope (SEM) technique was conducted to observe the distribution of nano-graphite particles within the concrete matrix. Debris collected from the compressive test was dried in heat oven at 110 °C for 24 hours before serving as SEM samples. Copper tape and gold coating were applied to samples to ensure good SEM images for non-conductive objects like concrete through both Secondary Electrons (SE) and Backscattered Electrons (BSE) detectors. A high-resolution, field-emission variable-pressure SEM machine was employed with 5-10 kV electrons energy.

3 Results and discussion

3.1 Compressive strength

The variations of compressive strength of concrete composites mixed with different proportions of nano-graphite materials are shown in Fig.6.

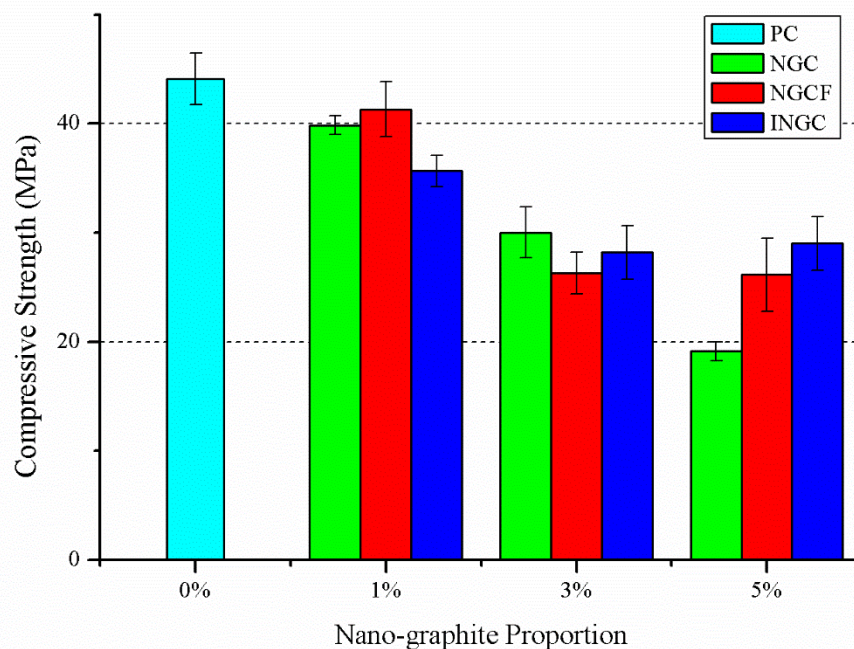


Fig.5 Compressive strength of different concrete composites at 28 days

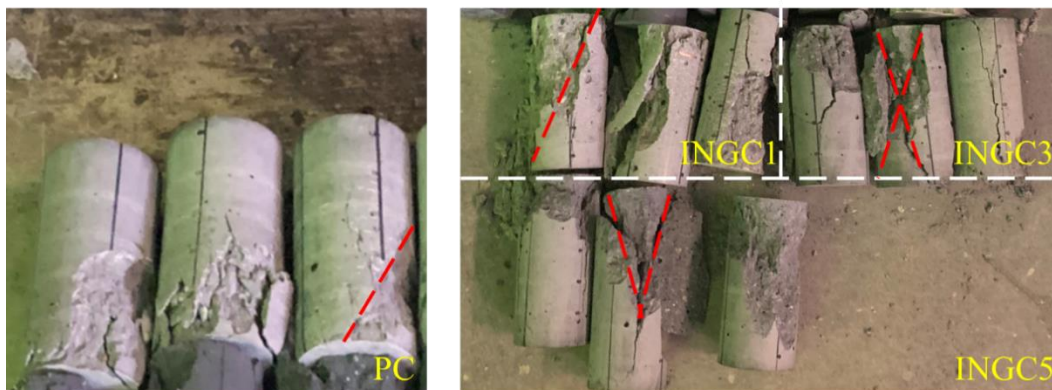
The bar at 0% indicates a 44.1 MPa average compressive strength of the plain concrete, which conforms to the standard design strength (42.5 MPa). According to Fig.6, the existence of nano-graphite materials has a negative impact on the compressive strength of concrete composites since all mixtures containing nano-graphite materials have lower compressive strengths than that of the plain concrete. This may be due to the lubrication effect of nano-graphite particles that would weaken the transition zone between aggregates and hydration products under the shearing strength, which determines the compressive strength of concrete [45].

Red and blue bars represent the compressive strength of concrete mixed with different nano-graphite materials at the same w/c ratio of 0.43 (NGCF and INGC). It is observed that the compressive strength decreases by respective amounts of 6.3% and

19.2% to 43.16 MPa and 35.65 MPa compared with the plain concrete, when 1 wt% of NG and ING were added. The compressive strength reduces further when the content of NG increases to 3 wt% dropping by 40.4% to 26.3 MPa compared with 44.1 MPa for the plain concrete. The increase of ING content to 3 wt% also results in a significant decrease of compressive strength to 28.2 MPa. However, it is evident that the use of ING at nano-graphite contents of 3 wt% and 5 wt% leads to little higher strengths than concrete composites mixed with NG. Meanwhile, it is assumed that there is a threshold content (3 wt% in this case) for the adverse effect of nano-graphite materials on the compressive strength of concrete composites. At these higher concentrations, the compressive strength of the concrete is more likely to be dominated by the strength of nano-graphite material itself that are distributed among the aggregate-cement transition zone.

Concrete composites containing NG with higher w/c ratio (green bars) show similar compressive strengths to other mixtures at the proportion of 1 wt% and 3 wt%, but much lower strengths at 5 wt%; this lower strength reflects the fact that a high w/c ratio has a significant negative influence on the concrete compressive strength due to the enhanced absorption effect arising from the higher content of pure nano-graphite.

Figure 7 shows some images of broken samples after the compressive strength testing. The failure modes conform to the traditional concrete failure theory [46], which is that concrete samples tend to break along the maximum shearing strength surface with the angle of $45^{\circ}\sim 60^{\circ}$ from the cross section under the uniaxial compressive loading. That is reflected in 'V' or '\ (slash)' shape of broken samples from the side view.



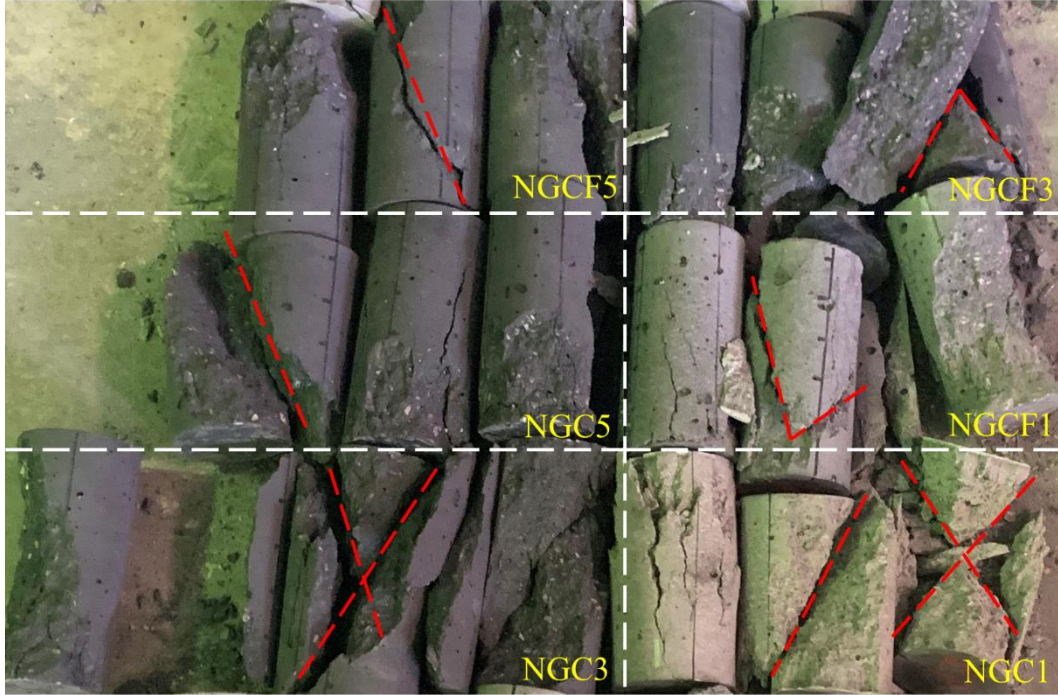


Fig.6 Failure modes of different concrete composites

3.2 Electrical resistivity

Figure 8 shows the electrical resistivity of different concrete composites mixed with and without nano-graphite materials at 28 days. Only concrete composites with the same w/c ratio of 0.43 are presented to assist identification of the influence on concrete electrical resistivity of different types of nano-graphite materials.

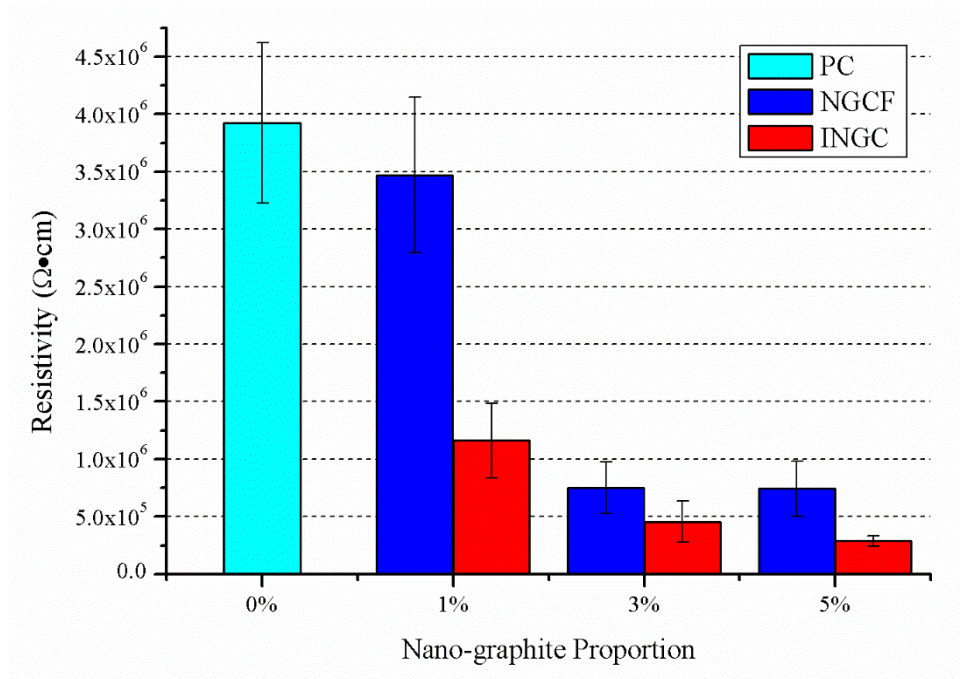


Fig.7 Electrical resistivity of different concrete composites at 28 days

There is a clear trend of reducing concrete electrical resistivity (and hence increasing conductivity) as the proportion of nano-graphite materials increases.

Furthermore, it is noticed that the effect of ING is much more significant than that of NG on reducing the electrical resistivity. By adding 1 wt% content of ING, the electrical resistivity drops by 70.3% compared with that of the plain concrete while there is only a 11.3% drop in electrical resistivity by using NG at the same proportion.

Figure 8 shows that the electrical resistivity of both NG-contained and ING-contained concretes drops further down significantly by 78.30% and 60.78% respectively when the nano-graphite content increases from 1 wt% to 3 wt%. According to the percolation theory in Fig.9 [47], nano-graphite particles tend to have more interaction with each other when the proportion of nano-graphite approaches the percolation threshold region. This bridging effect results in the extension of internal conductive paths shown as Fig.10, which is beneficial for electron transmission. When the content of nano-graphite material is over the percolation region, the electrical resistivity of concrete become stabilized as the bridging effect reaches saturated. That is why the concrete resistivity's reduction from 3% to 5% is much smaller than that from 1% to 3%. It is evident that the composite nano-graphite reaches its percolation threshold more quickly than the pure nano-graphite with less content of 1 wt%.

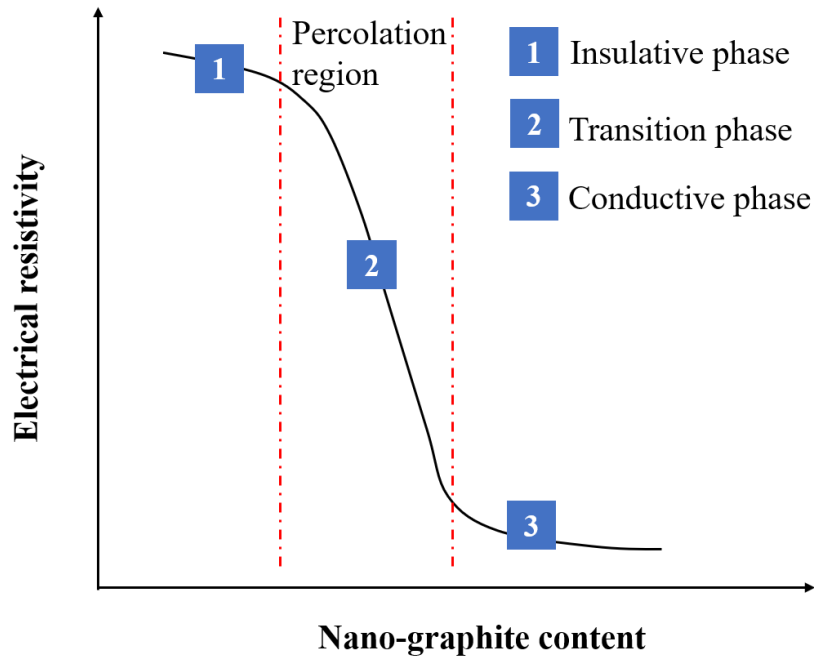


Fig.8 Schematic of the percolation theory

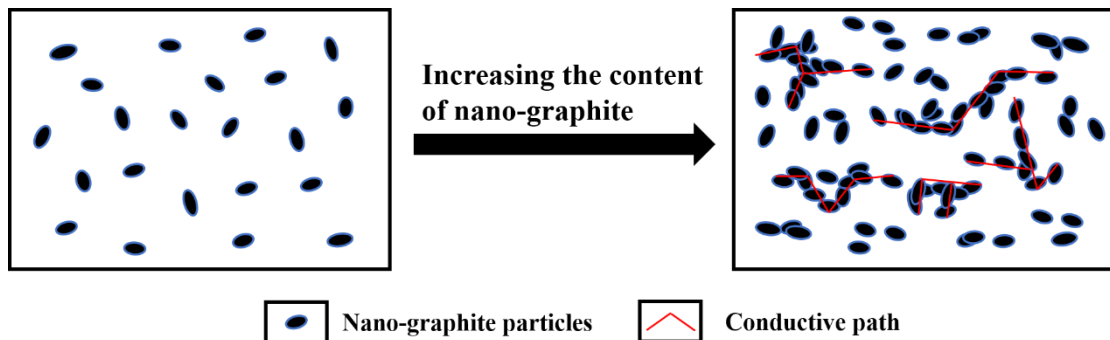


Fig.9 Schematic of the conductive model of nano-graphite particles

The electrical resistivity continues to fall with the increase of the proportion of nano-graphite materials. At the content of 3 wt% and 5 wt%, the electrical resistivity of concrete composites incorporating ING is almost half of that of NG-contained ones, respectively. The superiority of the composite nano-graphite in enhancing the concrete conductivity is mainly because this type of material contains a certain content of iron particles which obviously have more free electrons passing through conductive tunnels when connected to external power than normal nano-graphite [48].

3.3 SEM imaging

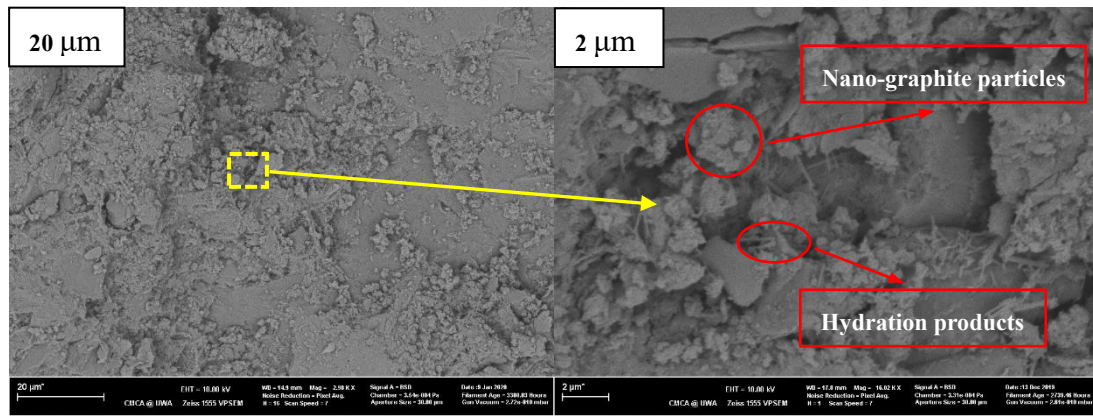
The distribution of nano-graphite materials within the concrete matrix was explored through the Scanning Electron Microscope (SEM) at scales of 20 μm and 2 μm with larger magnification.

From the high resolution (2 μm) images of different concrete composites with ING in Fig.11, it can be observed that ING particles tend to agglomerate and adhere to needle hydration products of the concrete matrix (Ettringite). The lower resolution (20 μm) images on Fig.11 show that the density and area of ING material on the concrete matrix increases with the increased ING content from 1 wt% to 5 wt%, which is also reflected in larger ING clusters at 2 μm . At 1 wt% and 3 wt%, the compressive strength of ettringite products are influenced by bonded formed with nano-graphite particles due to their lubrication effect. When the content of ING comes to 5 wt%, most ettringite products are wrapped by ING clusters and the strength of concrete is thus controlled by the strength of ING material, which links to the results shown in Fig.6. Besides, during the process of raising the content of nano-graphite, small ING agglomerations tend to group together and enhance the conductive effect on concrete composites.

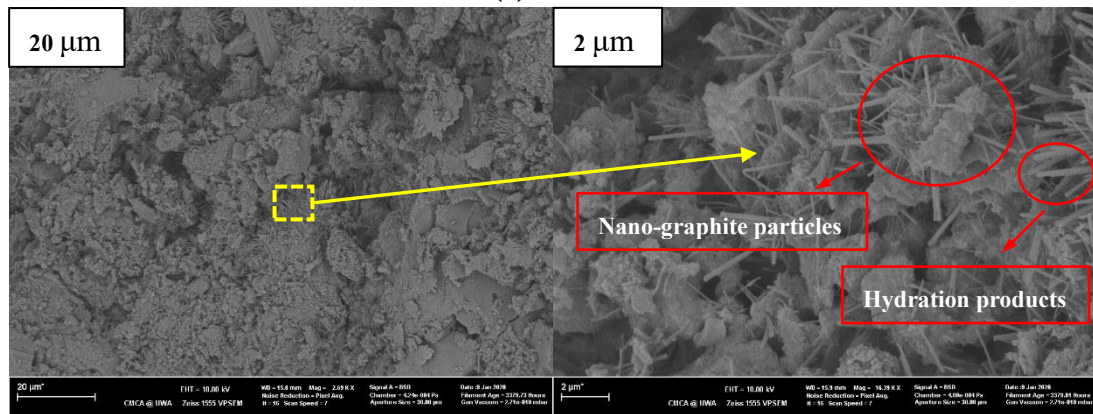
On the other hand, images at 2 μm scale in Fig.12 show that NG particles are in the form of platelets with larger specific surface area and disperse among those needle ettringite products.

At 1 wt% and 3 wt%, NG platelets are surrounded by ettringite products, and their lubrication effect may have an adverse impact on the concrete's shearing strength under external loading, which conforms to the results in Fig.6. However, it may be observed that most ettringite products are covered by NG platelets at the content of 5 wt% and the strength of NG material has a controlling influence on the concrete's compressive strength. The distribution of NG material observed within the concrete matrix at the scale of 20 μm in Fig. 12 also becomes even and dense with the increase of incorporation content from 1 wt% to 5 wt%. At 2 μm , Fig. 12 shows that more NG platelets start stacking up and overlapping together from 1 wt% to 5 wt%, which is beneficial for generating conductive paths that lead to a dramatic drop for the electrical resistivity of concrete composite as shown in Fig.8.

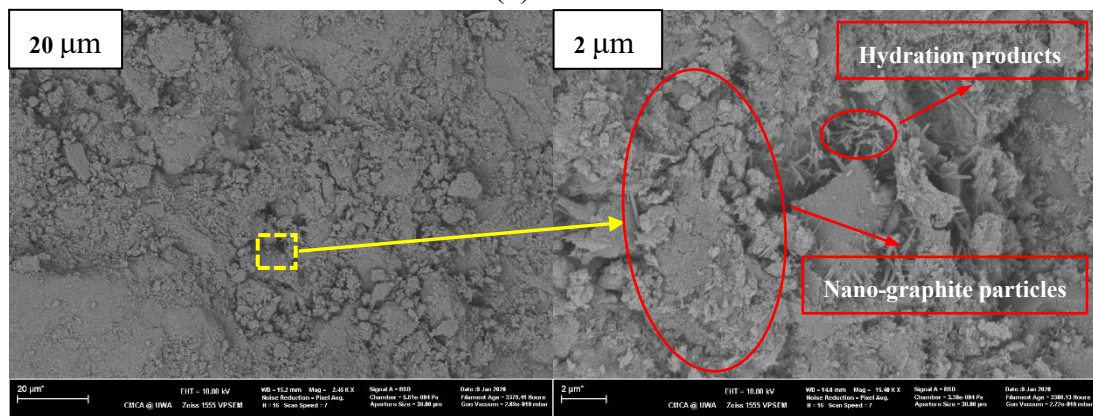
A comparison of images at 2 μm in Fig. 11 and Fig. 12 indicates that the existence pattern of ING particles within the concrete matrix makes it have greater interaction with ettringite products than NG at 1 wt%. Therefore, the composite nano-graphite shows a more significant impact on reducing the concrete's compressive strength than the pure nano-graphite at this content in Fig.6. This is also the probable reason why the concrete composites incorporating ING reaches its percolation threshold in electrical resistivity earlier than those mixed with NG.



(a) 1 wt%

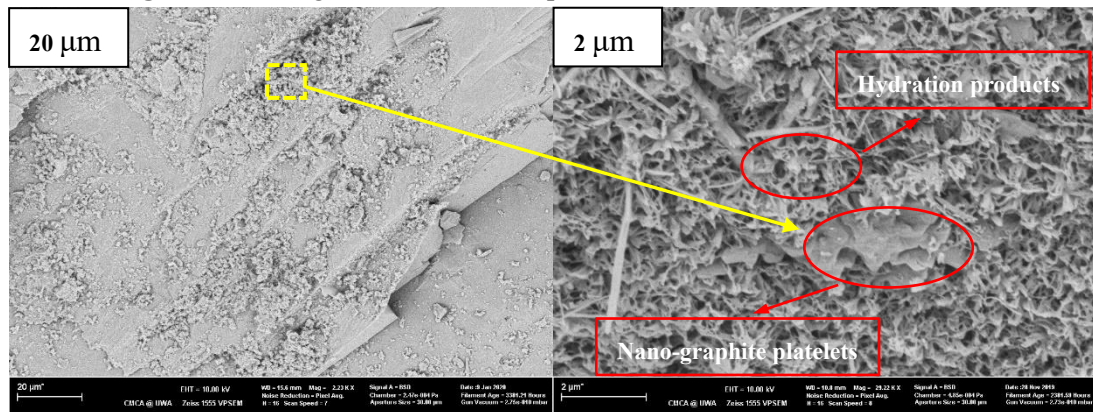


(b) 3 wt%



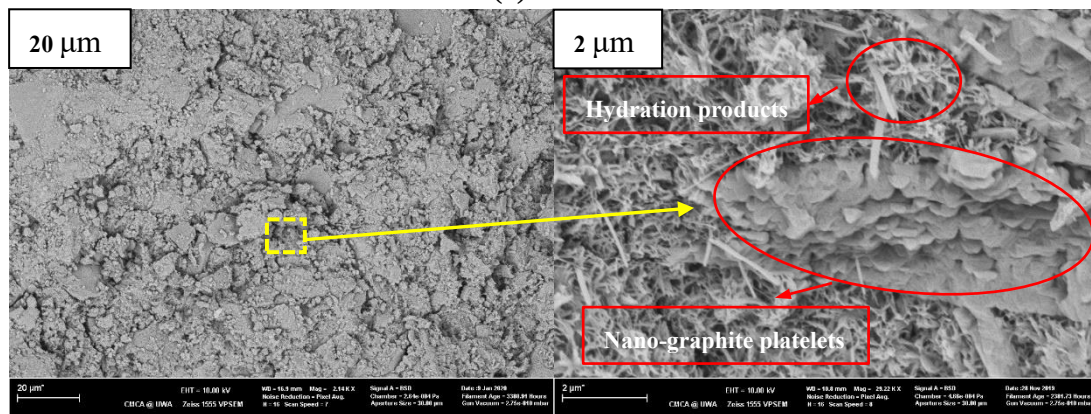
(c) 5 wt%

Fig.10 SEM images of concrete composites with different contents of ING



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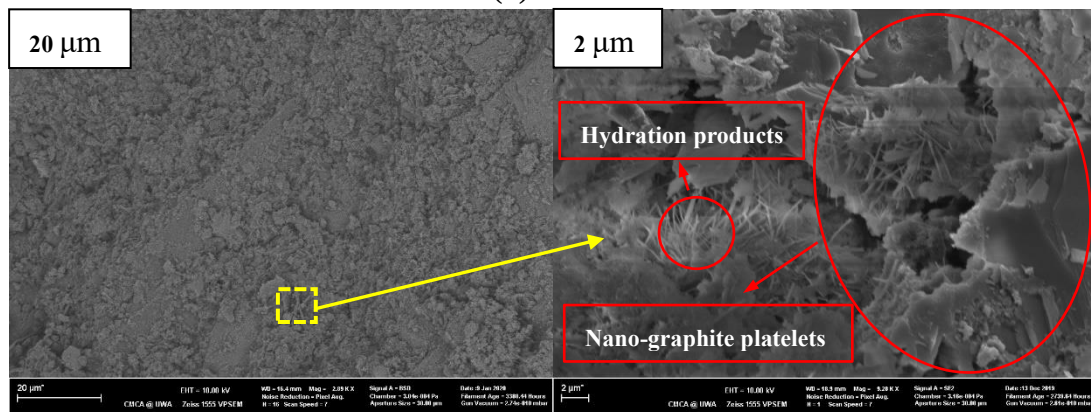
(a) 1 wt%



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(b) 3 wt%



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(c) 5 wt%

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4 Conclusions

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Fig.11 SEM images of concrete composites with different contents of NG

- In the present study, a hydrogen production by-product, which is a composite nano-graphite material that contains iron particles, was firstly explored on its effects on the electrical and mechanical properties of concrete composites, and compared with the plain concrete and concrete mixed with the pure nano-graphite. The potential mechanisms were studied through their microstructural characterization. From the experimental results, the relevant conclusions can be drawn as follows:
- 1) Both the pure and the composite nano-graphite materials have an adverse influence on the concrete's compressive strength because of the lubrication effect of nano-structural graphite.
 - 2) The w/c ratio of concrete mixtures could be held constant for different contents of the composite nano-graphite (ING) while the w/c ratio needed to be higher to ensure the workability of the mixing paste when the content of pure nano-graphite (NG) increased, which could also be achieved by adjusting the dosage of superplasticizer. This is because the larger BET specific surface area of NG leads to a greater absorption effect.
 - 3) Concrete composites mixed with 1 wt% content of ING show lower compressive strength than those using NG. But the use of ING at nano-graphite contents of 3 wt% and 5wt% leads to little higher concrete strengths than composites using NG.

- 4) There is a threshold content of 3 wt% for the adverse influence of nano-graphite materials on the concrete's compressive strength. After this content, the concrete's compressive strength is controlled by the strength of nano-graphite materials that are distributed in the transition zone between aggregates and hydration products.
- 5) Both nano-graphite materials can significantly enhance the electrical conductivity of concrete composites. The composite nano-graphite shows more effective in reducing concrete's electrical resistivity by reaching its percolation threshold sooner at the proportion of 1 wt% than the pure nano-graphite, which is attributed to the contained iron particles.
- 6) The composite nano-graphite particles show an obvious agglomeration effect around needle hydration products within the concrete matrix while the pure nano-graphite tends to spread among hydration products in formation of platelets. Increasing the dosage of nano-graphite materials from 1 wt% to 5 wt% promotes the overlapping of nano-graphite particles and extends conductive passages, which enhances the conductive effect.

Overall, from the preliminary study, this new nano-graphite by-product has a significant potential in producing conductive concrete for various applications compared with the normal nano-graphite. It can reach the effective concrete electrical conductivity by less mixing content, which is able to minimize its negative influence on the concrete's compressive strength. Besides, implementing this nano-graphite material into cementitious composites provides a beneficial way of recycling the industrial by-product of hydrogen production. Further experimental tests are required to identify the optimal mixing proportion for the composite nano-graphite and to combine with other cementitious composites such as high strength concrete to compensate the strength loss induced by nano-graphite.

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References

- [1] Shetty MS. Concrete Technology: Theory and Practice. India: S. chand,2008. ISBN: 9788121900034.
- [2] Majumder M, Gangopadhyay TK, Chakraborty AK, Dasgupta K, Bhattacharya DK. Fibre Bragg gratings in structural health monitoring—Present status and applications. *Sensors and Actuators A: Physical* 2008;147:150–64. <https://doi.org/10.1016/j.sna.2008.04.008>.
- [3] Imai M, Nakano R, Kono T, Ichinomiya T, Miura S, Mure M. Crack detection application for fiber reinforced concrete using BOCDA-based optical fiber strain sensor. *Journal of Structural*

- Engineering 2010;136:1001–8.
[https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0000195](https://doi.org/10.1061/(ASCE)ST.1943-541X.0000195).
- [4] Villalba S, Casas JR. Application of optical fiber distributed sensing to health monitoring of concrete structures. *Mechanical Systems and Signal Processing* 2013;39:441–51.
<https://doi.org/10.1016/j.ymssp.2012.01.027>.
- [5] Behnia A, Chai HK, Shiotani T. Advanced structural health monitoring of concrete structures with the aid of acoustic emission. *Construction and Building Materials* 2014;65:282–302.
<https://doi.org/10.1016/j.conbuildmat.2014.04.103>.
- [6] Goszczyńska B, Świt G, Trąmpczyński W, Krampikowska A, Tworzewska J, Tworzewski P. Experimental validation of concrete crack identification and location with acoustic emission method. *Archives of Civil and Mechanical Engineering* 2012;12:23–8.
<https://doi.org/10.1016/j.acme.2012.03.004>.
- [7] Hu S, Lu J, Xiao F. Evaluation of concrete fracture procedure based on acoustic emission parameters. *Construction and Building Materials* 2013;47:1249–56.
<https://doi.org/10.1016/j.conbuildmat.2013.06.034>.
- [8] Fujita Y, Mitani Y, Hamamoto Y. A method for crack detection on a concrete structure. 18th International Conference on Pattern Recognition (ICPR'06), 2006. p. 901–4.
<https://doi.org/10.1109/ICPR.2006.98>.
- [9] Yamaguchi T, Hashimoto S. Fast crack detection method for large-size concrete surface images using percolation-based image processing. *Machine Vision and Applications* 2010;21:797–809.
<https://doi.org/10.1007/s00138-009-0189-8>.
- [10] Cha Y-J, Choi W, Büyüköztürk O. Deep learning-based crack damage detection using convolutional neural networks. *Computer-Aided Civil and Infrastructure Engineering* 2017;32:361–78.
<https://doi.org/10.1111/mice.12263>.
- [11] Tian Z, Li Y, Zheng J, Wang S. A state-of-the-art on self-sensing concrete: Materials, fabrication and properties. *Composites Part B: Engineering* 2019;177:107437.
<https://doi.org/10.1016/j.compositesb.2019.107437>.
- [12] Liao W-I, Wang JX, Song G, Gu H, Olmi C, Mo YL, et al. Structural health monitoring of concrete columns subjected to seismic excitations using piezoceramic-based sensors. *Smart Materials and Structures* 2011;20:125015.
<https://doi.org/10.1088/0964-1726/20/12/125015>.
- [13] Meehan DG, Shoukai Wang, Chung DDL. Electrical-resistance-based sensing of impact damage in carbon fiber reinforced cement-based materials. *Journal of Intelligent Material Systems and Structures* 2010;21:83–105.
<https://doi.org/10.1177/1045389X09354786>.
- [14] Cañón A, Garcés P, Climent MA, Carmona J, Zornoza E. Feasibility of electrochemical chloride extraction from structural reinforced concrete using a sprayed conductive graphite powder–cement paste as anode. *Corrosion Science* 2013;77:128–34.
<https://doi.org/10.1016/j.corsci.2013.07.035>.
- [15] Minsk L. Electrically conductive asphalt for control of snow and ice accumulation. *High Research Record* 1968;227:57–63.
 URL:<https://trid.trb.org/view/110666>.
- [16] Caggiano MF, Bentley M. Route 130 bridge snowfree installation electrical analysis and

recommendations 1998.
 URL: <https://rosap.ntl.bts.gov/view/dot/15458>.

[17] Tuan CY. Roca Spur Bridge: The implementation of an innovative deicing technology. *Journal of Cold Regions Engineering* 2008;22:1–15.
[https://doi.org/10.1061/\(ASCE\)0887-381X\(2008\)22:1\(1\)](https://doi.org/10.1061/(ASCE)0887-381X(2008)22:1(1)).

[18] Derwin D, Booth P, Zaleski P, Marsey W, Flood W. Snowfree®, heated pavement system to eliminate icy runways, 2003.
<https://doi.org/10.4271/2003-01-2145>.

[19] El-Dieb AS, El-Ghareeb MA, Abdel-Rahman MAH, Nasr ESA. Multifunctional electrically conductive concrete using different fillers. *Journal of Building Engineering* 2018;15:61–9.
<https://doi.org/10.1016/j.jobbe.2017.10.012>.

[20] Wang L, Aslani F. A review on material design, performance, and practical application of electrically conductive cementitious composites. *Construction and Building Materials* 2019;229:116892.
<https://doi.org/10.1016/j.conbuildmat.2019.116892>.

[21] Banthia N, Djeridane S, Pigeon M. Electrical resistivity of carbon and steel micro-fiber reinforced cements. *Cement and Concrete Research* 1992;22:804–14.
[https://doi.org/10.1016/0008-8846\(92\)90104-4](https://doi.org/10.1016/0008-8846(92)90104-4).

[22] Yoo D-Y, Kim S, Lee SH. Self-sensing capability of ultra-high-performance concrete containing steel fibers and carbon nanotubes under tension. *Sensors and Actuators A: Physical* 2018;276:125–36.
<https://doi.org/10.1016/j.sna.2018.04.009>.

[23] Chung DDL. Strain sensors based on the electrical resistance change accompanying the reversible pull-out of conducting short fibers in a less conducting matrix. *Smart Materials and Structures* 1995;4:59–61.
<https://doi.org/10.1088/0964-1726/4/1/009>.

[24] Mustafa Sahmaran and Victor C. Li ML. Transport properties of engineered cementitious composites under chloride exposure. *ACI Materials Journal* n.d.;104.
<https://doi.org/10.14359/18964>.

[25] Al-Dahawi A, Sarwary MH, Öztürk O, Yıldırım G, Akın A, Şahmaran M, et al. Electrical percolation threshold of cementitious composites possessing self-sensing functionality incorporating different carbon-based materials. *Smart Materials and Structures* 2016;25:105005.
<https://doi.org/10.1088/0964-1726/25/10/105005>.

[26] Li H, Xiao H, Ou J. Effect of compressive strain on electrical resistivity of carbon black-filled cement-based composites. *Cement and Concrete Composites* 2006;28:824–8.
<https://doi.org/10.1016/j.cemconcomp.2006.05.004>.

[27] Sun S, Ding S, Han B, Dong S, Yu X, Zhou D, et al. Multi-layer graphene-engineered cementitious composites with multifunctionality/intelligence. *Composites Part B: Engineering* 2017;129:221–32.
<https://doi.org/10.1016/j.compositesb.2017.07.063>.

[28] Liantong M, Shaopeng W, Xiaoming L, Zheng C. Percolation model of graphite-modified asphalt concrete. *Journal of Wuhan University of Technology-Mater Sci Ed* 2005;20:111–3.
<https://doi.org/10.1007/BF02870889>.

[29] Dong W, Li W, Wang K, Luo Z, Sheng D. Self-sensing capabilities of cement-based sensor

with layer-distributed conductive rubber fibres. *Sensors and Actuators A: Physical* 2020;301:111763.
<https://doi.org/10.1016/j.sna.2019.111763>.

[30] Ahmed Sbia L, Peyvandi A, Soroushian P, Balachandra AM, Sobolev K. Evaluation of modified-graphite nanomaterials in concrete nanocomposite based on packing density principles. *Construction and Building Materials* 2015;76:413–22.
<https://doi.org/10.1016/j.conbuildmat.2014.12.019>.

[31] Han B, Yu X, Kwon E. A self-sensing carbon nanotube/cement composite for traffic monitoring. *Nanotechnology* 2009;20:445501.
<https://doi.org/10.1088/0957-4484/20/44/445501>.

[32] Meng W, Khayat KH. Effect of graphite nanoplatelets and carbon nanofibers on rheology, hydration, shrinkage, mechanical properties, and microstructure of UHPC. *Cement and Concrete Research* 2018;105:64–71.
<https://doi.org/10.1016/j.cemconres.2018.01.001>.

[33] Meng W, Khayat KH. Mechanical properties of ultra-high-performance concrete enhanced with graphite nanoplatelets and carbon nanofibers. *Composites Part B: Engineering* 2016;107:113–22.
<https://doi.org/10.1016/j.compositesb.2016.09.069>.

[34] Sun S, Han B, Jiang S, Yu X, Wang Y, Li H, et al. Nano graphite platelets-enabled piezoresistive cementitious composites for structural health monitoring. *Construction and Building Materials* 2017;136:314–28.
<https://doi.org/10.1016/j.conbuildmat.2017.01.006>.

[35] Han B, Ding S, Yu X. Intrinsic self-sensing concrete and structures: A review. *Measurement* 2015;59:110–28.
<https://doi.org/10.1016/j.measurement.2014.09.048>.

[36] Lee HK, Nam I-W, Tafesse M, Kim H-K. Fluctuation of electrical properties of carbon-based nanomaterials/cement composites: Case studies and parametric modeling. *Cement and Concrete Composites* 2019;102:55–70.
<https://doi.org/10.1016/j.cemconcomp.2019.04.008>.

[37] Xiang Q, Yu J. Graphene-based photocatalysts for hydrogen generation. *The Journal of Physical Chemistry Letters* 2013;4:753–9.
<https://doi.org/10.1021/jz302048d>.

[38] Standard, A., General purpose and blended cements, AS 3972-2010, Standard Australia. 2010. URL:<https://www.standards.org.au/standards-catalogue/sa-snz/manufacturing/bd-010/as--3972-2010>.

[39] Kosmatka ML, Wilson SH. Design and Control of Concrete Mixtures. Portland Cement Association, 16th edition, 2016. ISBN:978-0893122775.

[40] ASTM (American Society for Testing and Materials), C, Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory, ASTM C192/C192M-19. 2019. URL:<https://www.astm.org/Standards/C192>.

[41] Polder RB. Test methods for on site measurement of resistivity of concrete — a RILEM TC-154 technical recommendation. *Construction and Building Materials* 2001;15:125–31.
[https://doi.org/10.1016/S0950-0618\(00\)00061-1](https://doi.org/10.1016/S0950-0618(00)00061-1).

[42] Gao D, Sturm M, Mo YL. Electrical resistance of carbon-nanofiber concrete. *Smart Materials and Structures* 2009;18:095039.

544 <https://doi.org/10.1088/0964-1726/18/9/095039>

545 [43] Galao O, Baeza FJ, Zornoza E, Garcés P. Strain and damage sensing properties on
546 multifunctional cement composites with CNF admixture. *Cement and Concrete Composites*
547 2014;46:90–8.
548 <https://doi.org/10.1016/j.cemconcomp.2013.11.009>

549 [44] ASTM (American Society for Testing and Materials), C, Standard Test Method for
550 Compressive Strength of Cylindrical Concrete Specimens, ASTM C39/C39M-20. 2020.
551 URL:<https://www.astm.org/Standards/C39.htm>.

552 [45] Cui X, Sun S, Han B, Yu X, Ouyang J, Zeng S, et al. Mechanical, thermal and electromagnetic
553 properties of nanographite platelets modified cementitious composites. *Composites Part A: Applied*
554 *Science and Manufacturing* 2017;93:49–58.
555 <https://doi.org/10.1016/j.compositesa.2016.11.017>.

556 [46] Xiao J, Li J, Zhang Ch. Mechanical properties of recycled aggregate concrete under uniaxial
557 loading. *Cement and Concrete Research* 2005;35:1187–94.
558 <https://doi.org/10.1016/j.cemconres.2004.09.020>.

559 [47] Chung DDL. Piezoresistive cement-based materials for strain sensing. *Journal of Intelligent*
560 *Material Systems and Structures* 2002;13:599–609.
561 <https://doi.org/10.1106/104538902031861>.

562 [48] Pietronero L, Strässler S, Zeller HR, Rice MJ. Electrical conductivity of a graphite layer.
563 *Physical Review B* 1980;22:904–10.
564 <https://doi.org/10.1103/PhysRevB.22.904>.